Important Questions - Unitwise

1. Define "Semi-structured data" and provide two common file format examples.
2. How does "Unstructured data" differ from "Structured data" in terms of searchability?
3. State the formal definition of Big Data according to Gartner's 3Vs.
4. Briefly explain the different types of Analytics and explain their significance.
5. Define Apache Hadoop and name its two primary core components.
6. State the fundamental difference between RDBMS and Hadoop in terms of "Schema-on-Write" vs. "Schema-on-Read."
7. Briefly explain the roles of the NameNode and the DataNode in an HDFS cluster.
8. Mention one major architectural difference between Hadoop 1.x and Hadoop 2.x (e.g., the introduction of YARN).
9. Name three popular commercial distributions of Hadoop available in the market.
10. Define "Data Locality" and its importance in reducing network bandwidth usage.
11. Name any two tools in the Hadoop Ecosystem used for data ingestion (e.g., Flume, Sqoop).
12. Define MapReduce and state its primary purpose in a Hadoop cluster.
13. What is the fundamental role of a Mapper in a MapReduce job?
14. Describe the primary task performed by the Reducer.
15. What is a "Combiner," and why is it often referred to as a "Mini-Reducer"?
16. What is the main responsibility of the Partitioner during the MapReduce workflow?
17. Is it possible to run a MapReduce job with zero reducers? If so, what is the output format?

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1. Explain how a Data Warehouse and a Big Data environment (like Hadoop) can work together in a single organization.
2. Compare Traditional Business Intelligence and Big Data Analytics based on "Data Volume" and "Processing Complexity."

3. Differentiate between "Predictive Analytics" and "Prescriptive Analytics" using a real-world business scenario.
4. Discuss two technological drivers (e.g., Cloud, IoT) that caused the "Sudden Hype" around Big Data Analytics in the last decade.
5. Discuss why traditional systems fail to process Big Data, focusing on the concepts of "Vertical Scaling" vs. "Horizontal Scaling."
6. Explain the "Cost-Effectiveness" and "Scalability" advantages of Hadoop compared to traditional high-end storage area networks (SAN).
7. Describe the challenge of "Partial Failure" in distributed systems and how Hadoop handles it.
8. Explain the concept of a "Block" in HDFS. Why is the default block size (e.g., 128MB) much larger than traditional file system blocks?
9. Briefly describe the purpose of **Hive** and **Pig** within the Hadoop ecosystem and how they simplify MapReduce programming.
10. Briefly outline the process of how a file is written to HDFS (mentioning the pipeline of DataNodes).
11. Discuss the introduction of "Erasure Coding" in Hadoop 3.x and how it differs from traditional replication.
12. What is the specific purpose of the Secondary NameNode? (Note: Clarify that it is *not* a backup/failover for the primary NameNode).
13. Outline the step-by-step lifecycle of a MapReduce job, from InputFormat to OutputFormat.
14. Compare and contrast the Combiner and the Reducer. Are there any scenarios where a Combiner cannot be used?
15. Explain how MapReduce leverages the "Data Locality" principle to optimize processing speed.
16. Explain the significance of YARN?
17. Explain the importance of the Writable and WritableComparable interfaces in MapReduce programming.